

Chapter 1

Rings

1.1 Rings and Subrings

We now continue to generalize the familiar arithmetic operations of addition and multiplication. Just as with groups, results from the algebra of arithmetic will be extended to a host of non-numerical objects, e.g., matrices, functions, polynomials, etc.

Definition 1. A *ring* is a nonempty set, say R , with two binary operations (usually written as multiplication and addition)¹ that satisfies the following:

1. If $a, b \in R$ then $a + b \in R$. (closed under addition)
2. $a + (b + c) = (a + b) + c$ (associative addition)
3. $a + b = b + a$ (commutative addition)
4. There exists an 0_R such that $a + 0_R = 0_R + a = a$ (additive identity)
5. There exists an $-a \in R$ such that $a + (-a) = 0_R$ for every $a \in R$. (additive inverse)
6. If $a, b \in R$ then $ab \in R$. (closed under multiplication)
7. $a(bc) = (ab)c$ (associative multiplication)
8. $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ (distribution holds).

- Note that parts 1-5 say that a ring is an *abelian group*.²

Proposition 1. If a and b are two elements in R then $(-a)(-b) = ab$.

Proof. By definition $a + (-a) = 0$. $(a + (-a))(-b) = 0$ implies $a(-b) + (-a)(-b) = 0$ by distribution. The additive inverses of $a(-b)$ is unique thus $a(-b) + (-a)(-b) = 0$ implies that $(-a)(-b) = ab$. $\square$

- In particular for rings like $\mathbb{Z}$ and $\mathbb{R}$, the product of two negatives is a positive. Note the importance of distribution.

Definition 2. A *commutative ring* is a ring R such that $ab = ba$ for all $a, b \in R$.

¹But of course the operations are not necessarily addition and multiplication. $+$ and $\times$; $\oplus$ and $\otimes$; are among the different notations that may be used in varying texts.

²Commutative groups are also called abelian in honor of Norwegian mathematician Niels Abel.

Definition 3. A ring with *identity*³ is a ring R with an element 1_R such that $a1_R = 1_Ra$ for all $a \in R$.

- It is important to check that a potential 1_R is both a *right identity*, $a1_R = a$; and a *left identity*, $1_Rb = b$.

Theorem 1. *If a ring has an identity, it is unique.*

Proof. Suppose to the contrary that a ring R has two identity elements 1 and $1'$. $a1 = a$ for all $a \in R$ including $1' \in R$. Thus $1' \cdot 1 = 1'$ and $1 \cdot 1' = 1$. By the definition of a ring $1' = 1 \cdot 1' = 1' \cdot 1 = 1$. Therefore $1' = 1$, and the multiplicative identity is unique. $\square$

1.2 Subrings

Definition 4. Let R be a ring and S be a subset of R .⁴ If S is a ring under the same operations as R , then S is a *subring*.

The following two theorems provide shortcuts when trying to show that a subset is a subring. Particularly, The second theorem can save quite a bit of work.

Theorem 2. *Suppose that R is a ring and $S \subseteq R$ such that the following hold:*

1. *If $a, b \in S$ then $a + b \in S$ (closed under addition)*
2. *If $a, b \in S$ then $ab \in S$ (closed under multiplication)*
3. $0_R \in S$.
4. *If $a \in S$ then there is an $-a \in S$ such that $a + (-a) = 0_R$.*

Then S is a subring of R .

Proof. Parts 2, 3, 7, and 8 of the Ring definition hold for all elements in R since it is a ring. So they hold for all elements in S . The remaining requirements 1-4 in the theorem satisfy the remaining requirements. $\square$

- Rings and thus subrings must be nonempty. A requirement which is covered by 3. Since $0_R \in S \Rightarrow S \neq \emptyset$.

Theorem 3. *A nonempty subset S of a ring R is a subring if whenever $a, b \in S$:*

1. $ab \in S$ (closed under multiplication)
2. $a - b \in S$ (closed under subtraction)

Proof. If $a - b \in S$ for all $a, b \in S$ then S is a subgroup of R (subgroup test), and since R is additively commutative it follows that S is an abelian subgroup of R under addition (this also means that $0_R \in S$). S is multiplicatively and additively associative; and S is distributive over addition because these properties hold in R . $\square$

³The term *unity* is often used in place of identity. A ring with an identity element is said to “have unity”.

⁴Notated: $S \subseteq R$.

1.3 Integral Domains and Fields

For $\mathbb{Z}_n$ it is easy to see that $ab = 0$ does *not* imply that a or b is 0. For example, in $\mathbb{Z}_6$ $2 \cdot 3 \bmod 6 = 6 \bmod 6 = 0$. Of course many rings (indeed the most familiar rings) do have this property. In $\mathbb{R}$, if $x \cdot y = 0$ then $x = 0$ or $y = 0$.

Definition 5. An *integral domain* is a commutative ring R with identity such that $1_R \neq 0_R$ ⁵ and if $ab = 0_R$ then $a = 0_R$ or $b = 0_R$ for any $a, b \in R$.⁶

Theorem 4. Let R be an integral domain and let $a, b, c \in R$ where $c \neq 0$. If $ac = bc$, then $a = b$.

Proof. Since rings have distribution, $ac = bc \Rightarrow ac - bc = 0 \Rightarrow (a - b)c = 0$. Because R is an integral domain $a - b = 0$ which implies that $a = b$. $\square$

Note that we could not prove theorem 4 using division, i.e., $\frac{ac}{c} = \frac{bc}{c}$, because we cannot assume that $c^{-1} \in R$. For example, $3x = 3y \Rightarrow x = y$ in $\mathbb{Z}$ but $\frac{1}{3} \notin \mathbb{Z}$. While $\mathbb{Z}$ is an integral domain, it has no element x such that $3x = 1$. That is, 3 (and all other numbers except 1) does not have a multiplicative inverse. All the elements in $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ (except 0) do have multiplicative inverses. Thus $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are all examples of the following definition.

Definition 6. A *field* is a commutative ring R with identity $1_R \neq 0_R$ ⁷ if for every $a \neq 0_R$ in R , there exists an $a^{-1} \in R$ such that $aa^{-1} = 1_R$.

- Sets $\supseteq$ Groups $\supseteq$ Rings $\supseteq$ Commutative Rings with identity $\supseteq$ Integral Domains $\supseteq$ Fields

Theorem 5. $\mathbb{Z}_p$ is a field if p is a prime.

Proof. This is left as an exercise. $\square$

Theorem 6. Cancellation. Let a, b, c be elements in an integral domain. Suppose that $a \neq 0$. Then $ab = ac \Rightarrow b = c$.

Proof. Let $a \neq 0$ for elements a, b, c in an integral domain, R . $ab = ac \Rightarrow a(b - c) = 0_R$. Because R is an integral domain either $a = 0$ or $b - c = 0$. By assumption, $a \neq 0$ so then $b - c = 0 \Rightarrow b = c$. $\square$

Theorem 7. A field is an integral domain.⁸

Proof. This is left as an exercise. $\square$

Theorem 8. A finite integral domain is a field.

Proof. Let R be a finite integral domain with identity. Let $a \in R$ and $a \neq 0_R$. All we need to show is that a has an inverse. If $a = 1_R$ then $a^{-1} = 1_R$. Now suppose that $a \neq 1_R$ (if no such exists then we are done). Since R is finite there are two integers such that $i < j$ and $a^i = a^j$ (otherwise R would be infinite). By cancellation (above theorem) we have that $a^{j-i} = 1_R$. Because $a \neq 1_R$ we know that $j - i > 1$. Thus $a^{j-i-1}(a) = 1_R$ as needed. $\square$

⁵The condition $1_R \neq 0_R$ is necessary so that the ring $\{0\}$ is not an integral domain.

⁶If a, b are nonzero elements in a ring and $ab = 0$ then a and b are called *zero divisors*. Thus an integral domain is a commutative ring with identity that has no zero divisors.

⁷Once again, this assumption is needed so that $\{0\}$ is excluded from this class.

⁸Often a *field* is defined as an integral domain in which every nonzero element has an multiplicative inverse.